Further internal research on DML v53: rank promises and finite measurements

Research working note prepared for author review

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Abstract

We give independent analytic extensions of the supplied minimax argument. For a state-rank promise at most s, the exact coefficient is $\frac{d}{d-1}\binom{2k}{k}4^{-k}$, where $k = \min(s, \lfloor d/2 \rfloor)$. The same real-two-frame measurement is optimal for every such promise. For pure inputs the coefficient is $d/[2(d-1)]$. We also give a finite-nonattainment proof for all $d \geq 3$ and all these rank promises, and determine the positive-guarantee outcome threshold for pure inputs in dimension four. These are internal derivations, not established claims of literature novelty. External primary-source and priority verification remains incomplete.

1 Setting and the promise

Fix the conjugation basis in dimension $d \geq 2$. Use the unnormalized trace norm and

$$\mathcal{M}(\rho) = \frac{1}{2}\|\rho - \rho^*\|_1, \qquad \mathrm{TV}[p, q] = \frac{1}{2}\|p - q\|_1.$$

The measurement is chosen before the unknown input, and may have continuous outcomes. This definition makes no assumption that conjugation is a physically implementable channel. For $1 \leq s \leq d$, define

$$\alpha_s(\mathsf{E}) = \inf_{\substack{\operatorname{rank}\rho \leq s \\ \mathcal{M}(\rho) > 0}} \frac{\mathrm{TV}[p_{\mathsf{E}}(\rho), p_{\mathsf{E}}(\rho^*)]}{\mathcal{M}(\rho)}, \qquad \beta_{d,s} = \sup_{\mathsf{E}} \alpha_s(\mathsf{E}).$$

Let $r = \lfloor d/2 \rfloor$ and $k = \min(s, r)$. The rank promise is strict. A purity value alone is not the same promise.

Proposition 1 (Exactly which directions are allowed)**.** *Up to nonzero positive scaling, the directions $\rho - \rho^*$ with $\operatorname{rank}\rho \leq s$ are precisely the nonzero matrices*

$$X = X^\dagger, \quad X^* = -X, \quad \operatorname{rank} X \leq 2k.$$

Proof. The forward implication follows from $\operatorname{rank}(\rho - \rho^*) \leq 2\operatorname{rank}\rho$ and the paired spectrum of an imaginary Hermitian matrix. Conversely, write the positive and negative parts of X as X_+, X_-. Conjugation exchanges them, and $\operatorname{Tr} X_+ = \operatorname{Tr} X_- > 0$. Thus

$$\rho = \frac{X_+}{\operatorname{Tr} X_+}, \qquad \rho^* = \frac{X_-}{\operatorname{Tr} X_+}, \qquad \rho - \rho^* = \frac{X}{\operatorname{Tr} X_+}.$$

The rank of ρ is $\operatorname{rank} X/2 \leq k \leq s$. Its imaginarity is one. Hence every allowed direction is realized by an actual state within the promise. □

2 An exact rank-sensitive minimax law

Define

$$c_{d-1} = \frac{\Gamma((d-1)/2)}{\sqrt{\pi}\,\Gamma(d/2)}, \qquad \mu_{d,k} = \frac{\Gamma(k+1/2)\Gamma(d/2)}{\Gamma(k)\Gamma((d+1)/2)}.$$

For a uniform unit real vector u, $\mu_{d,k}$ is the mean length of its projection onto a fixed $2k$-plane. The formula also covers $2k = d$, when this length is one.

Theorem 1 (Rank-promise coefficient)**.** *For every $d \geq 2$ and $1 \leq s \leq d$,*

$$\boxed{\beta_{d,s} = \frac{d}{d-1}\frac{\Gamma(k+1/2)}{\sqrt{\pi}\,\Gamma(k+1)} = \frac{d}{d-1}\frac{\binom{2k}{k}}{4^k}, \quad k = \min(s, \lfloor d/2 \rfloor).} \tag{1}$$

An attaining POVM, independent of s, is

$$\mathsf{E}_{\mathrm{fr}}(du\,dv) = d|\psi_{u,v}\rangle\langle\psi_{u,v}|\,d\nu(u,v), \qquad \psi_{u,v} = \frac{u+iv}{\sqrt{2}},$$

where ν is invariant probability on ordered orthonormal real two-frames. A worst direction has exactly k equal nonzero skew singular-value pairs.

Proof. First consider the invariant orbit of a real skew matrix A_0 with k singular-value pairs equal to one and all remaining singular values zero. Put $A = OA_0O^T$ for Haar $O \in O(d)$. For fixed real $y \neq 0$, $Ay \perp y$ and

$$\mathbb{E}\|Ay\| = \mu_{d,k}\|y\|.$$

Invariance under the stabilizer of y implies that, conditional on its length, the direction of Ay is uniform in $y^\perp$. Consequently

$$\mathbb{E}_A|x^TAy| = c_{d-1}\mu_{d,k}\|P_{y^\perp}x\|\,\|y\|. \tag{2}$$

The zero-vector case contributes zero. For a unit complex vector $\psi = x + iy$ this gives

$$\mathbb{E}_A|\langle\psi|iA|\psi\rangle| = 2c_{d-1}\mu_{d,k}\sqrt{\|x\|^2\|y\|^2 - (x\cdot y)^2} \leq c_{d-1}\mu_{d,k}.$$

Spectral decomposition and the triangle inequality extend the bound to every positive effect F, with right side $c_{d-1}\mu_{d,k}\operatorname{Tr}F$. Write an arbitrary POVM as $F_\omega d\lambda(\omega)$ and integrate $\operatorname{Tr}F_\omega$ to d. Since $\|iA\|_1 = 2k$ and every orbit direction is allowed by the rank promise,

$$\alpha_s(\mathsf{E}) \leq \mathbb{E}_A\frac{\int|\operatorname{Tr}(F_\omega iA)|\,d\lambda}{2k} \leq \frac{d\,c_{d-1}\mu_{d,k}}{2k}.$$

The last expression simplifies to Eq. (1).

For the lower bound, orthogonal invariance and reversal $v \mapsto -v$ give $\int|\psi_{u,v}\rangle\langle\psi_{u,v}|d\nu = \mathbb{I}/d$, so the frame POVM is normalized. Put an allowed $X = iA$ in canonical form with nonnegative paired singular values $a_1, \ldots, a_k$, padding zeros if necessary. Its response is

$$R_{\mathrm{fr}}(X) = \frac{d\,c_{d-1}}{2\sum_j a_j}\mathbb{E}_u\sqrt{\sum_{j=1}^{k} a_j^2(u_{2j-1}^2 + u_{2j}^2)}.$$

At fixed $\sum_j a_j$, the expectation is convex in (a_j) and symmetric under their permutations. Averaging all permutations therefore shows that it is minimized by equal a_j. For that spectrum the expectation is $a\mu_{d,k}$, giving the same upper bound. This proves attainability and the worst-spectrum statement, including lower-rank directions. □

Corollary 1 (Pure inputs and recovery of v53)**.** *For pure unknown inputs,*

$$\beta_{d,1} = \frac{d}{2(d-1)} \longrightarrow \frac{1}{2}. \tag{3}$$

The frame measurement has this ratio for every nonreal pure input. Taking $s \geq r$ in Eq. (1) recovers the even/odd dimension formula in v53. For fixed s and $d \to \infty$ the coefficient approaches $\binom{2s}{s}/4^s$; when k also grows it behaves as $d/[(d-1)\sqrt{\pi k}]$.

Proof. For pure $\rho = |\psi\rangle\langle\psi|$, the nonzero conjugation direction has rank two, so its skew spectrum has one pair. The remaining assertions follow directly from the gamma identities and Stirling's formula. In particular the unrestricted dimension decay comes from admitting a growing number of nonzero skew pairs. □

Proposition 2 (Complex Haar baseline under the same promise)**.** *For $\mathsf{U}(d\psi) = d|\psi\rangle\langle\psi|d\mu(\psi)$ with complex Haar probability,*

$$\alpha_s(\mathsf{U}) = \frac{\Gamma(k+1/2)}{\sqrt{\pi}\,\Gamma(k+1)} = \frac{d-1}{d}\beta_{d,s}.$$

Proof. The response is a convex symmetric function of the k nonnegative paired eigenvalues at fixed sum. At equal eigenvalues let p_+, p_- be the probabilities in the two k-dimensional eigenspaces. With $T = p_+ + p_-$ and $U = p_+/T$, one has $\mathbb{E}T = 2k/d$ and an independent $U \sim \mathrm{Beta}(k,k)$. The normalized response is

$$\frac{d}{2k}\mathbb{E}[T|2U-1|] = \mathbb{E}|2U-1| = \frac{\Gamma(k+1/2)}{\sqrt{\pi}\,\Gamma(k+1)}.$$

□

Thus the constant-order behavior for a fixed-rank promise is already present in the Haar baseline. The improvement over that baseline is the finite-dimensional factor $d/(d-1)$; no new asymptotic scaling advantage over Haar is asserted.

3 No finite measurement attains these optima

Theorem 2 (All-rank finite nonattainment)**.** *For $d \geq 3$, every finite-outcome POVM obeys $\alpha_s(\mathsf{E}) < \beta_{d,s}$ for each $1 \leq s \leq d$. For every fixed outcome budget m, the supremum over POVMs with at most m outcomes is also strictly below $\beta_{d,s}$.*

Proof. Suppose a finite POVM attained the optimum. Refine its effects into their finitely many rank-one spectral components. Classical coarse-graining contracts total variation, so this refinement still attains the optimum. Write its nonzero effects as $w_j|\psi_j\rangle\langle\psi_j|$, with $\sum_j w_j = d$.

The orbital average used above is now exact:

$$\mathbb{E}_A R_{\mathsf{E}}(iA) = \frac{\beta_{d,s}}{d}\sum_j w_j m_j, \qquad m_j = 2\sqrt{\|x_j\|^2\|y_j\|^2 - (x_j\cdot y_j)^2} \leq 1.$$

The assumed lower bound $R_{\mathsf{E}}(iA) \geq \beta_{d,s}$ forces $m_j = 1$ for every active effect and forces the continuous response to be constant on the entire flat-rank-$2k$ orbit. For any chosen effect this means, after a real basis change, $\psi_j = (e_1 + ie_2)/\sqrt{2}$.

Choose A_0 with a skew block on coordinates $(1,3)$; if $k \geq 2$ also put a block on $(2,4)$, then put the remaining $k-2$ blocks on $(5,6),(7,8),\ldots$. For $k = 1$ all other coordinates are zero. This is available for every $d \geq 3$, $k \leq \lfloor d/2 \rfloor$. Let R_θ rotate coordinates $(2,3)$ and set $A_\theta = R_\theta A_0 R_\theta^T$. The response of the chosen effect is proportional to

$$|e_1^T A_\theta e_2| = |\sin\theta|.$$

It has a strictly positive jump in its first derivative at zero. Every other term is a positive multiple of the absolute value of a real analytic function. At a zero its derivative jump is nonnegative, and away from a zero its derivative is continuous. Hence the finite sum has a positive derivative jump and cannot be constant. This contradicts orbital constancy.

Finally, the set of m-outcome POVMs, allowing zero effects, is compact. The coefficient is a continuous minimum over the compact set $\{X \in \mathcal{V}_d : \|X\|_1 = 1, \operatorname{rank} X \leq 2k\}$. A fixed-budget supremum is therefore attained; since equality is impossible, it is strictly smaller than $\beta_{d,s}$. □

This does not give a quantitative dependence of the gap on m. Dimension two is an exception: a sharp Y measurement attains one.

4 A changed outcome threshold in dimension four

For unrestricted inputs v53 requires $d(d-1)/2 + 1$ outcomes for a positive coefficient. The following promise changes that threshold.

Theorem 3 (Pure inputs in dimension four)**.** *The minimum number of outcomes with $\alpha_1(\mathsf{E}) > 0$ in dimension four is exactly four. An explicit four-outcome POVM has $\alpha_1 = 1/\sqrt{6}$. For unrestricted inputs the threshold is seven. The value $1/\sqrt{6}$ is a feasible guarantee, not a claimed optimal four-outcome coefficient for pure inputs.*

Proof. The space of real skew 4×4 matrices has dimension six. Any m-outcome readout has rank at most $m-1$ on this space. For $m \leq 3$, its kernel has dimension at least four. The Pfaffian quadratic form

$$q(A) = A_{12}A_{34} - A_{13}A_{24} + A_{14}A_{23}$$

has signature $(3, 3)$. A subspace on which it has no nonzero zero must be positive or negative definite and has dimension at most three. Thus the kernel contains a nonzero A with $q(A) = 0$. Such an A has rank two and corresponds to a pure-state direction. The coefficient is zero, proving the lower bound on outcomes.

For an explicit construction use the real quaternion matrices

$$L_1 = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix},$$

$$L_3 = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

They satisfy $L_a^T = -L_a$ and $L_aL_b + L_bL_a = -2\delta_{ab}\mathbb{I}$. Let $H_a = iL_a$ and let n_j be the four unit tetrahedral vertices $(1,1,1), (1,-1,-1), (-1,1,-1), (-1,-1,1)$ divided by $\sqrt{3}$. Set

$$E_j = \frac{\mathbb{I} + n_j \cdot H}{4}, \qquad j = 1, \ldots, 4.$$

These are rank-two positive effects and sum to $\mathbb{I}$. For a rank-two direction $X = ia(uv^T - vu^T)$, with u, v real orthonormal, the three vectors L_au form an orthonormal basis of $u^\perp$. Thus $z_a = -v^TL_au$ is a unit vector, and

$$R_{\mathsf{E}}(X) = \frac{1}{4}\sum_{j=1}^{4} |n_j \cdot z|.$$

Every $z \in S^2$ is attainable, and the tetrahedral absolute sum has minimum $2\sqrt{2/3}$ (evaluate the normals to pairs of its generators). Consequently $\alpha_1(\mathsf{E}) = 1/\sqrt{6} > 0$. □

The same measurement is blind to the rank-two state

$$\rho_{\text{blind}} = \frac{\mathbb{I} + iR_1}{4}, \qquad R_1 = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}.$$

Indeed $\mathrm{Tr}(H_a i R_1) = 0$, so both conjugate inputs give uniform outcome probabilities. The state has $\mathcal{M}(\rho_{\text{blind}}) = 1$. This verifies why the pure-input guarantee cannot be applied to arbitrary mixed inputs.

5 Approximate promises and research interpretation

Proposition 3 (A robust consequence with an explicit limitation)**.** *Suppose there is a state σ of rank at most s with $\|\rho - \sigma\|_1 \le \eta$. For the frame measurement,*

$$\mathrm{TV}[p(\rho), p(\rho^*)] \ge \beta_{d,s}\mathcal{M}(\rho) - (1 + \beta_{d,s})\eta.$$

If $\mathcal{M}(\rho) > 0$, divide by it to obtain a ratio bound. There is no uniform small ratio error from this argument when $\mathcal{M}(\rho)$ is arbitrarily small.

Proof. Contraction and the triangle inequality bound the change in the two-input total variation by η. The distance-to-real-state function $\mathcal{M}$ is one-Lipschitz in trace norm. Apply the exact rank-promise guarantee to σ and use $\mathcal{M}(\sigma) \ge \mathcal{M}(\rho) - \eta$. □

The rank formula, the common optimal measurement, and the strict gap for a finite outcome budget suggest studying how the rank promise changes measurement design. These extensions use orthogonal averaging, convexity, and measurement norms, methods already present in the surrounding literature. The Haar comparison gives the same fixed-rank asymptotic order. Their publication value therefore still depends on primary-source and priority verification. No claim that a literature search found these statements absent is made here.

6 Checks actually performed

The supplied Python script checks exact rational values against beta quadrature for $2 \le d \le 40$, recovers the v53 parity formula, and performs 21 real-frame Monte Carlo cases with 100000 samples each. Seven direct density-matrix probability cases check that the promised directions are realized by positive rank-k states. The four-dimensional effects, their guarantee, a blind mixed state, and 63 flat-orbit cusp curves are also checked. Saved results and implementation versions accompany the note. These computations check consistency and examples; they do not replace the analytic proofs or establish literature novelty.

Primary literature access was attempted through the configured proxy. Both arXiv and Crossref returned a forbidden tunnel response. Required research domains have been saved in an environment configuration draft, which does not itself apply or publish the network change. Full-text verification remains pending.

Assistance record. This research note, its analytic extensions, and verification code were developed with ChatGPT assistance. They require independent author and expert review before being incorporated into a submission. DML v53 has been preserved.